

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 898+780

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

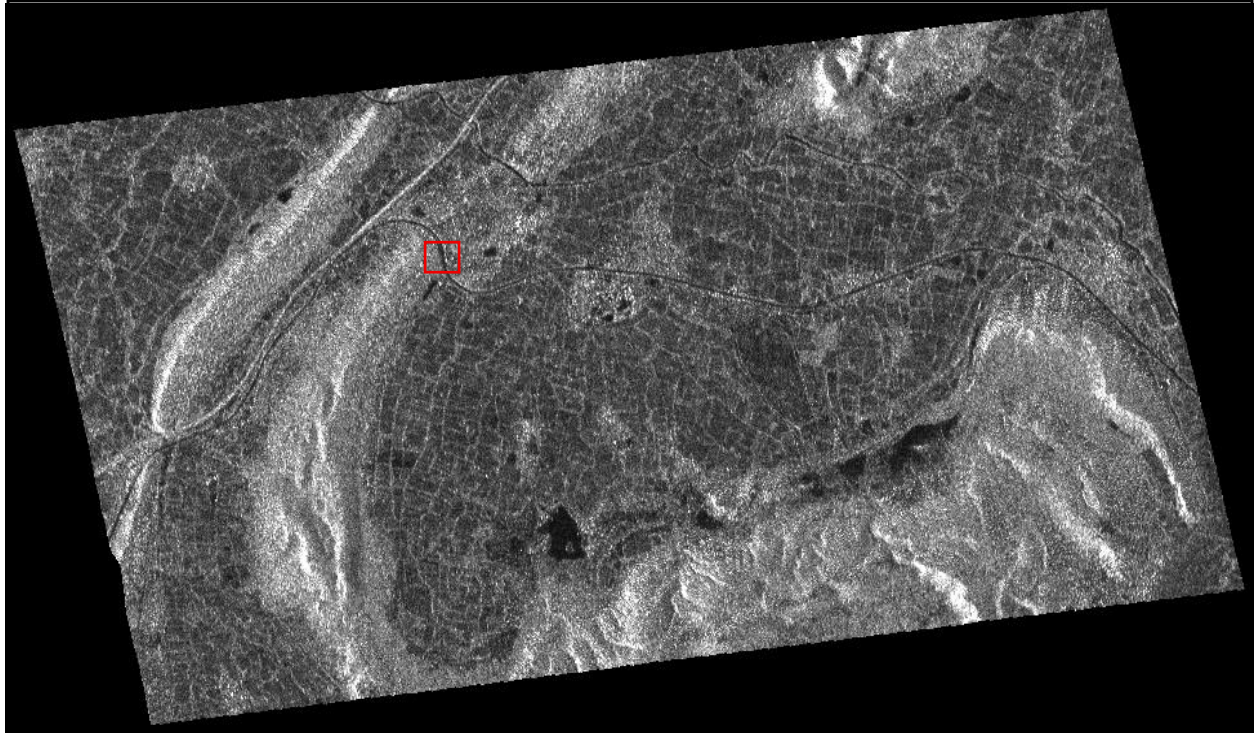
Flight Direction - DESCENDING

Polarization – VV+VH

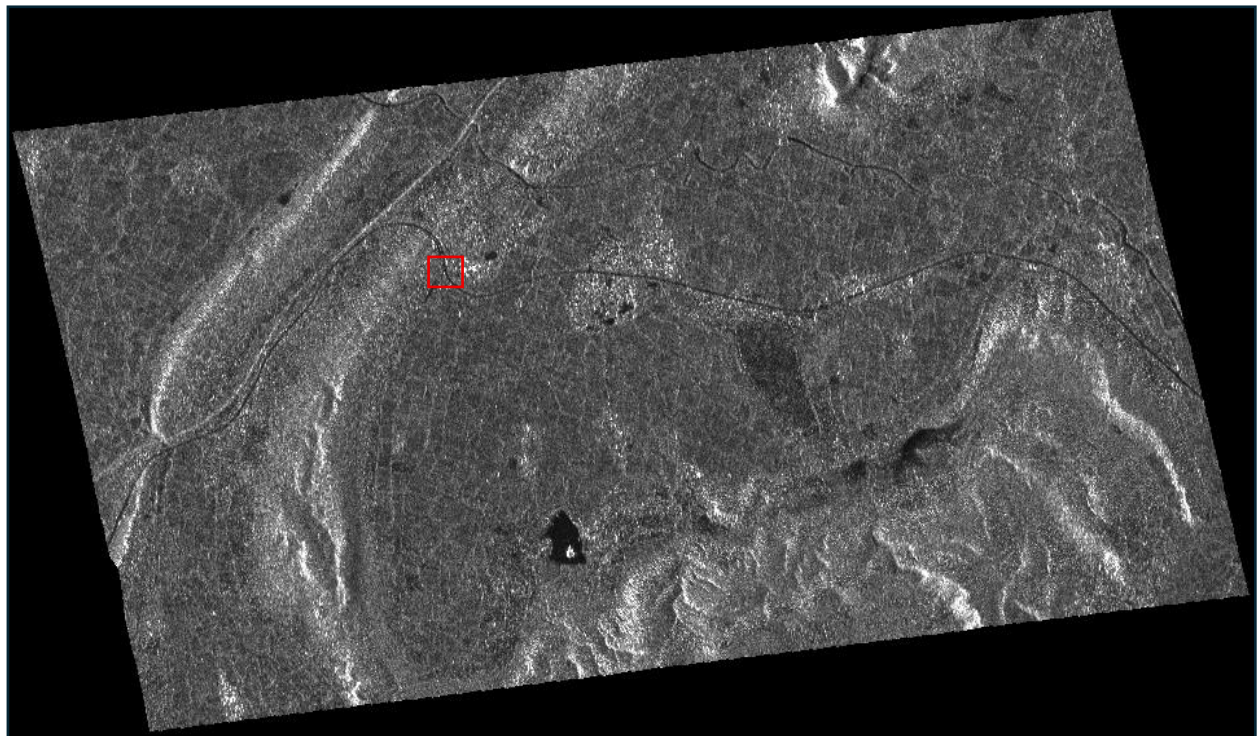
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

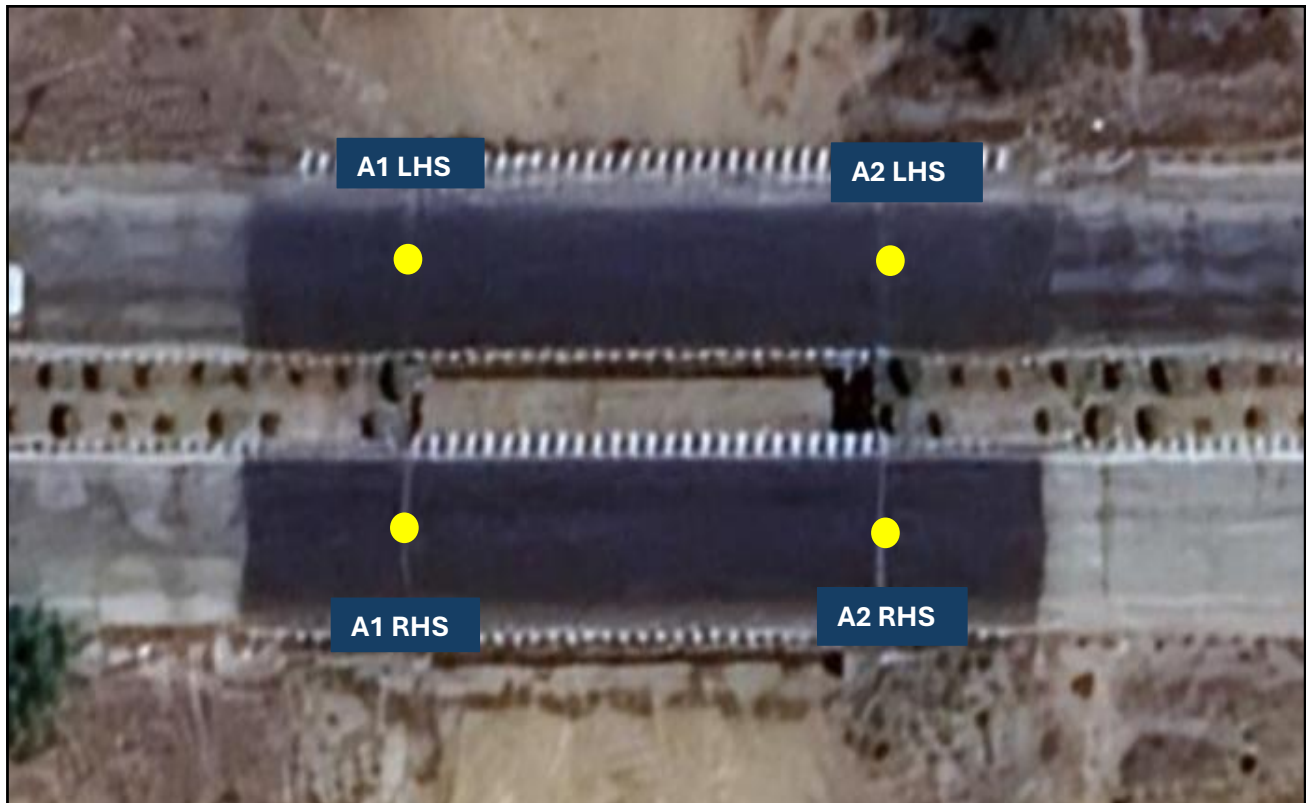


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

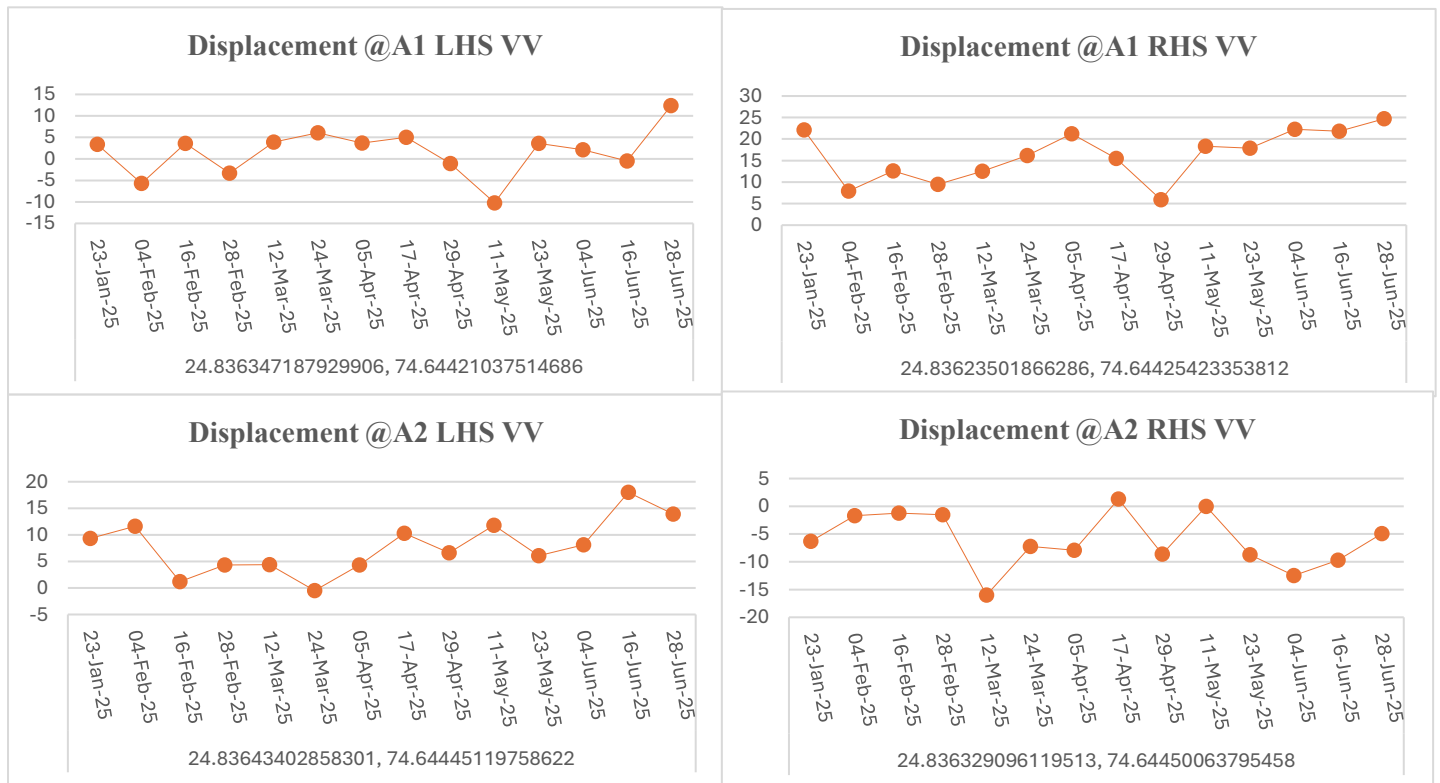
MNB – CH-898+780

Length – 27.11M

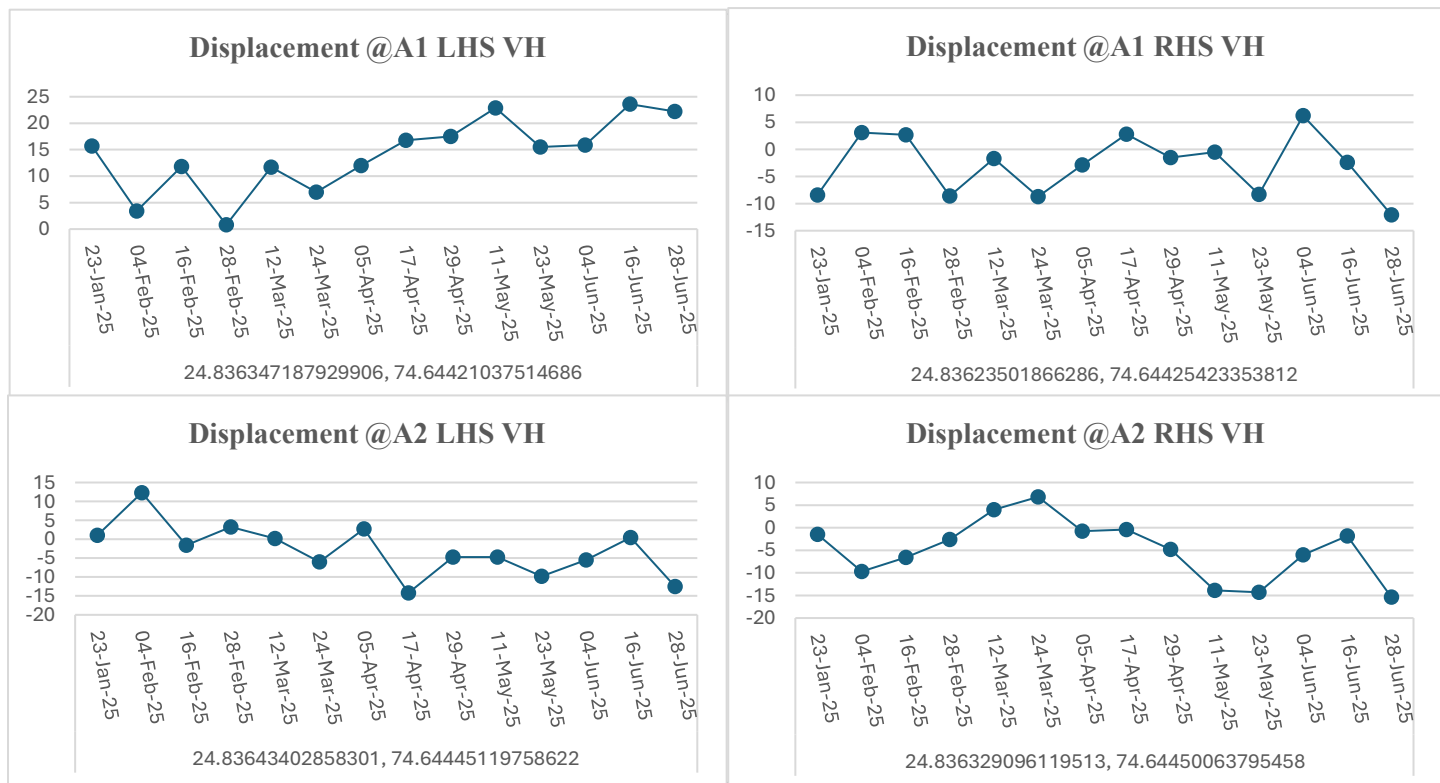


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

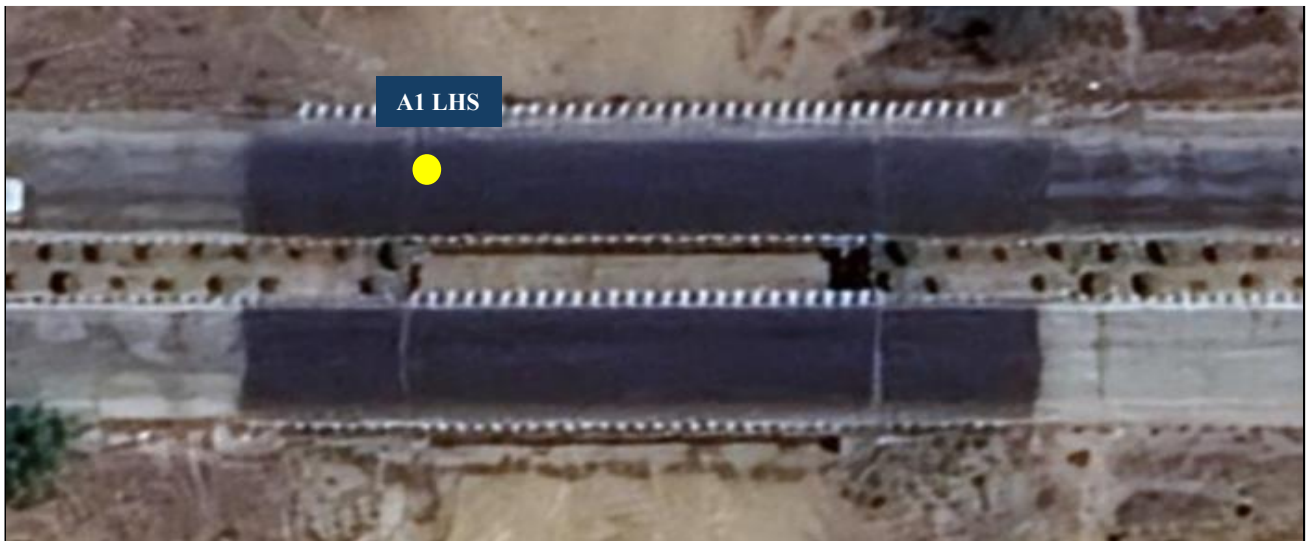
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



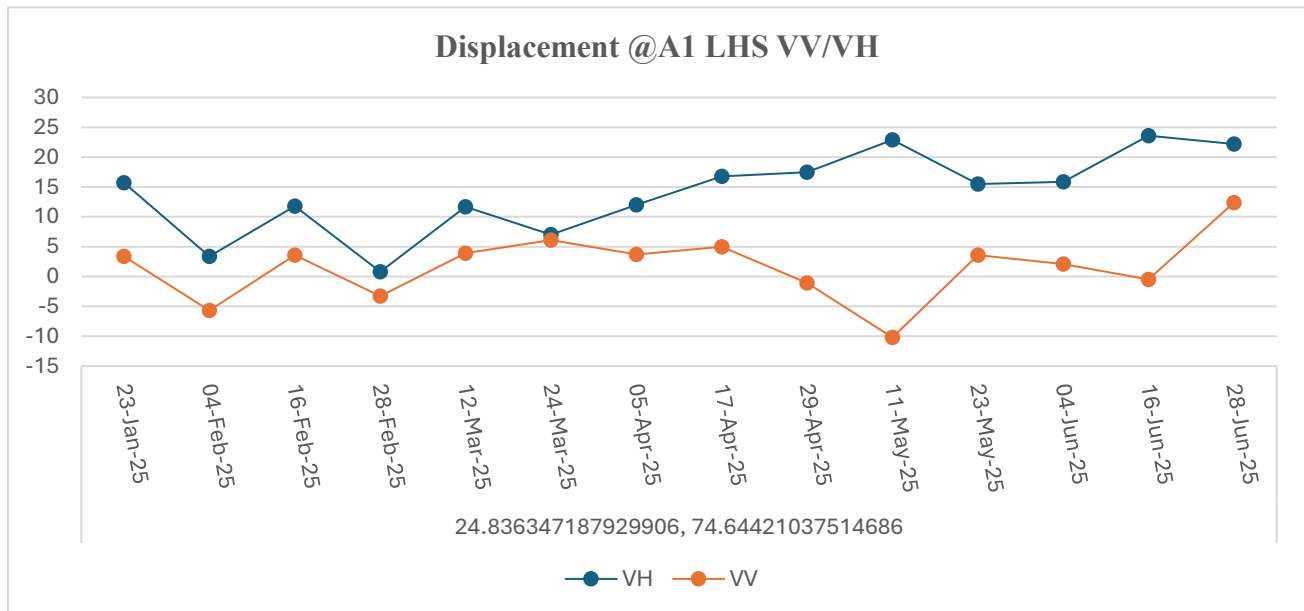
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.836347187929906, 74.64421037514686)



Key Observations:

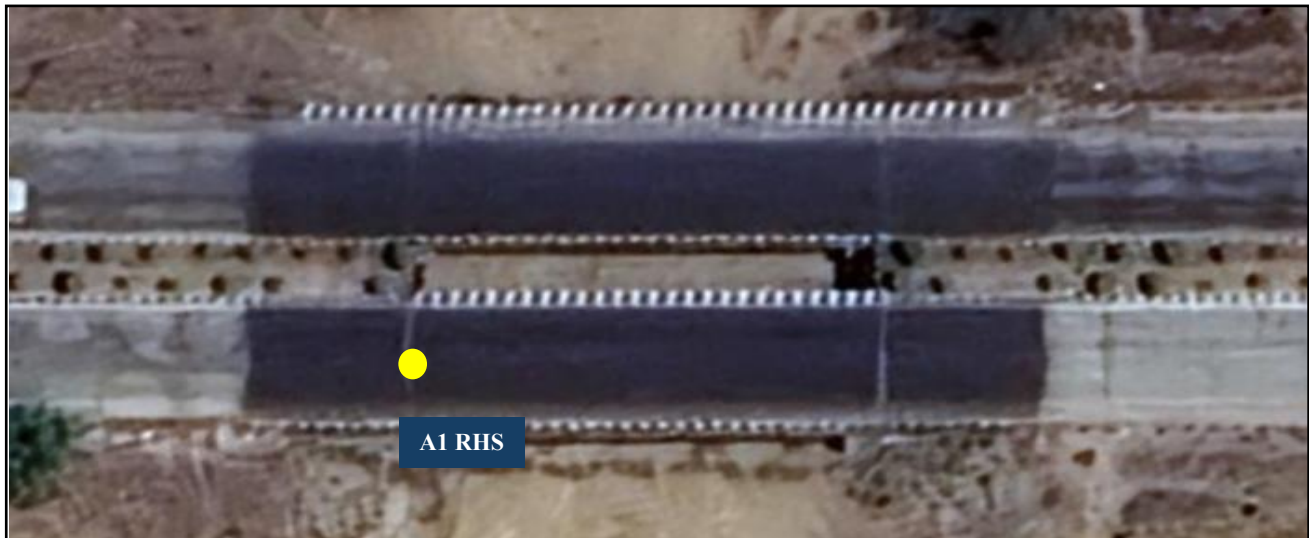
Displacement Magnitude Patterns:

- Vertical Displacement (VV)** Vertical displacement shows moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from –10 mm to +12 mm.
- Horizontal Displacement (VH)** Horizontal displacement exhibits a strong and predominantly positive trend, ranging roughly between +5 mm and +24 mm.

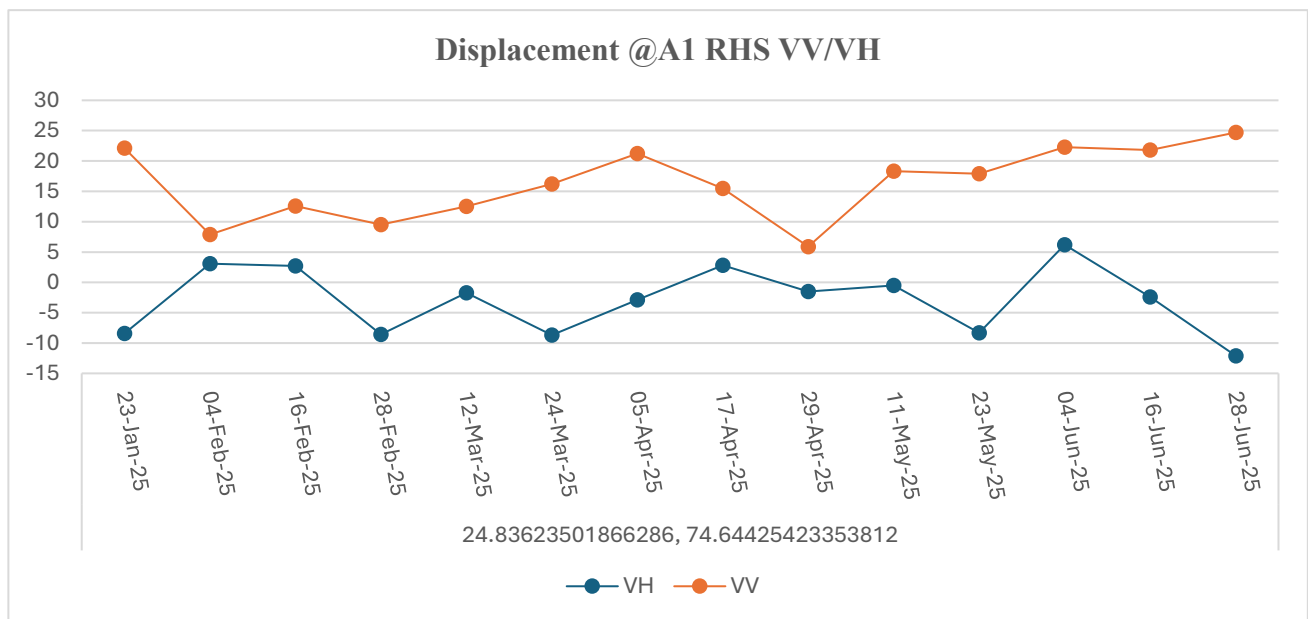
Critical Events & Anomalies:

- March 2025:** VH increases steadily to +12–15 mm, while VV fluctuates near +3–5 mm, indicating lateral displacement dominance with limited vertical participation.
- June 2025:** VV rises sharply to +12 mm, while VH remains high (+22 mm), marking a combined lateral displacement and vertical uplift intensification toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.83623501866286, 74.64425423353812)



Key Observations:

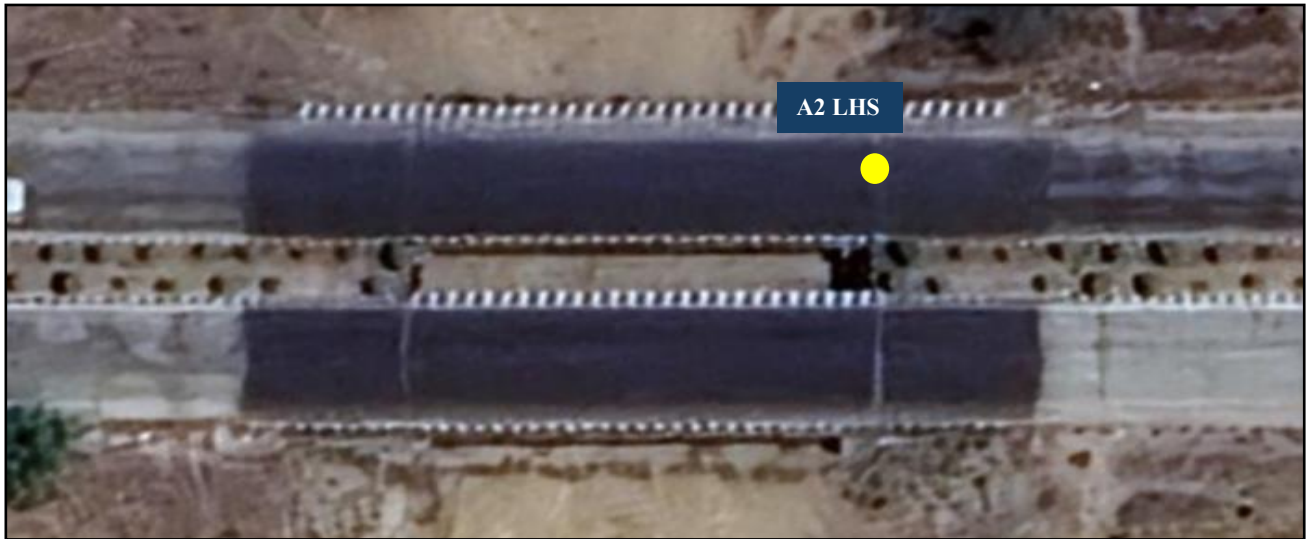
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate variability with predominantly negative values, ranging roughly between +5 mm and –12 mm.
- **Vertical displacement (VV)** Vertical displacement remains consistently positive throughout the monitoring period (Jan–Jun 2025), indicating sustained vertical uplift at the A1 RHS location.

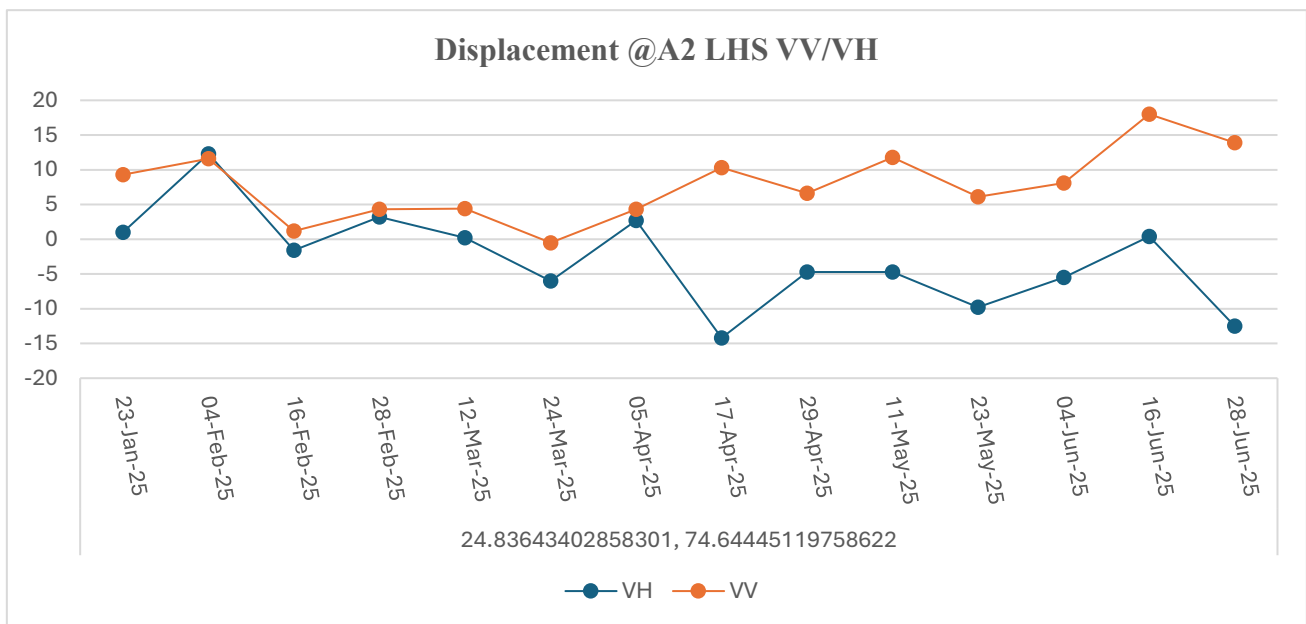
Critical Events & Anomalies:

- **January 2025:** VH starts at –9 mm, while VV is already high (+22 mm), indicating early vertical uplift occurring alongside lateral settlement.
- **May 2025:** VV rebounds sharply to +18 mm, while VH drops again to –9 mm, marking a highly asymmetric deformation phase.
- **June 2025:** VH briefly turns positive (+6 mm), while VV increases to +22 mm, representing a short-lived coupled uplift and lateral recovery event.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.83643402858301, 74.64445119758622)



Key Observations:

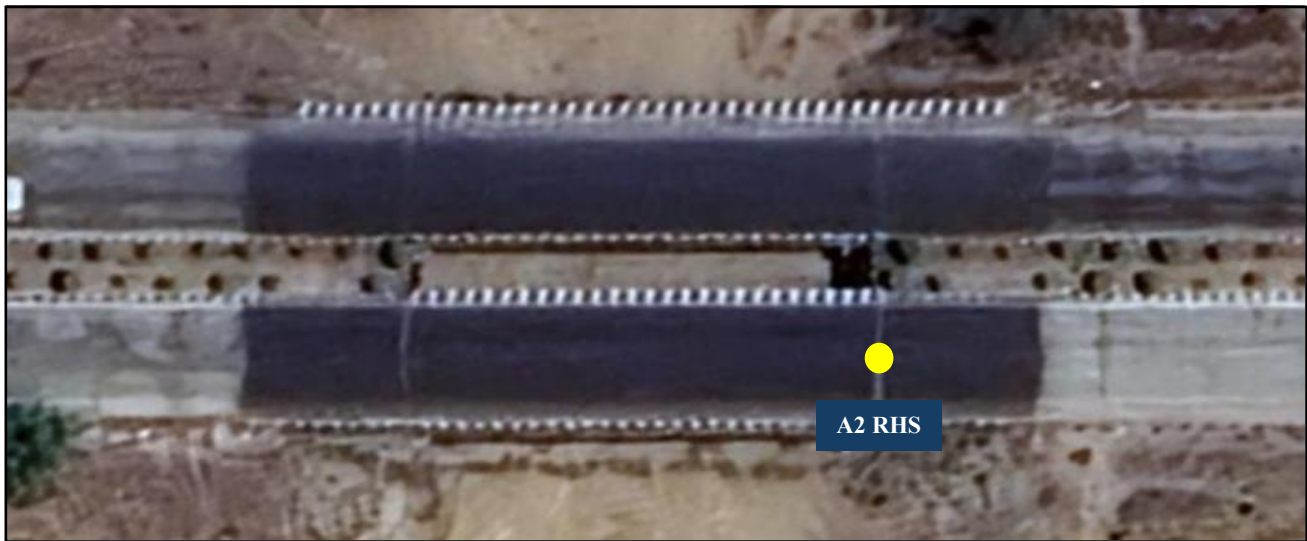
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits strong variability with predominantly negative values, ranging roughly between +12 mm and -15 mm.
- **Vertical Displacement (VV)** Vertical displacement shows moderate variability with a predominantly positive trend over the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A2 LHS location.

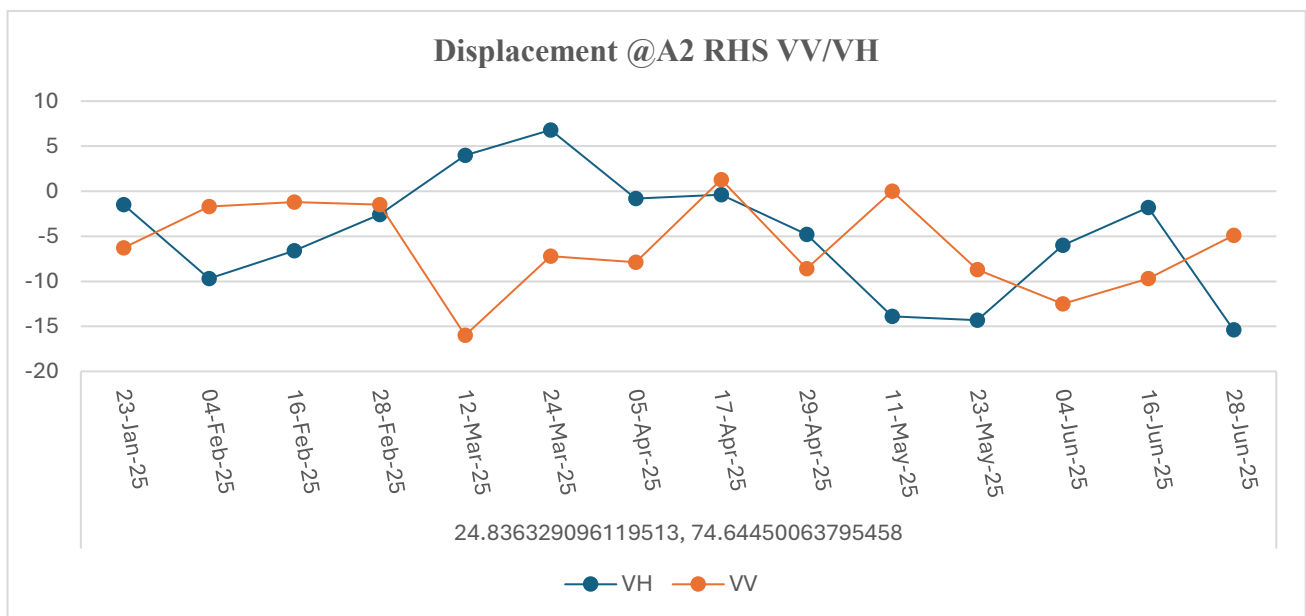
Critical Events & Anomalies:

- **April 2025:** VH recovers to +4 mm, while VV increases to +5 mm, representing a short-term coupled recovery phase.
- **June 2025:** VV increases further to +14–18 mm, while VH fluctuates between -5 mm and +3 mm, indicating continued vertical uplift with unresolved lateral instability toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.836329096119513, 74.64450063795458)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits high variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from +1 mm to –15 mm.
- **Horizontal (VH) displacement** Horizontal displacement shows moderate variability with a predominantly negative trend, ranging roughly between +7 mm and –16 mm.

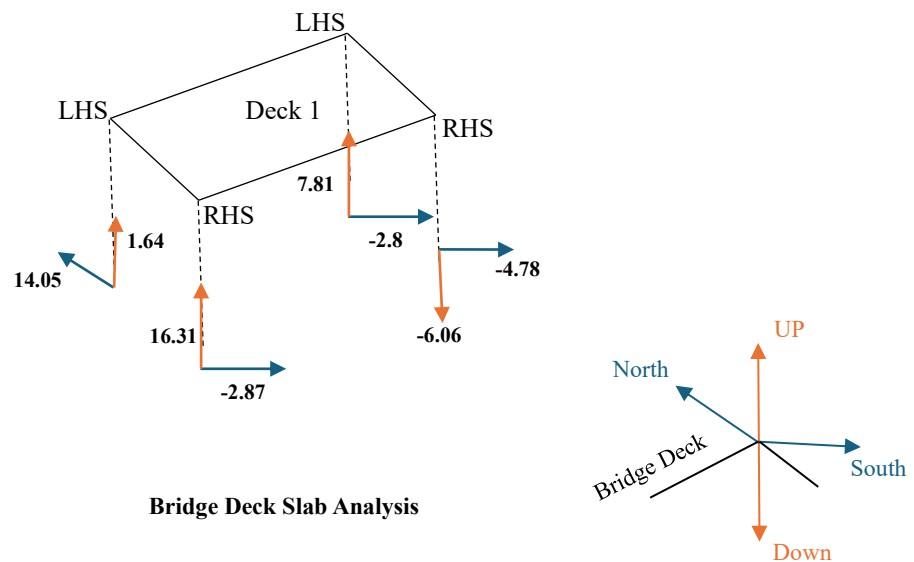
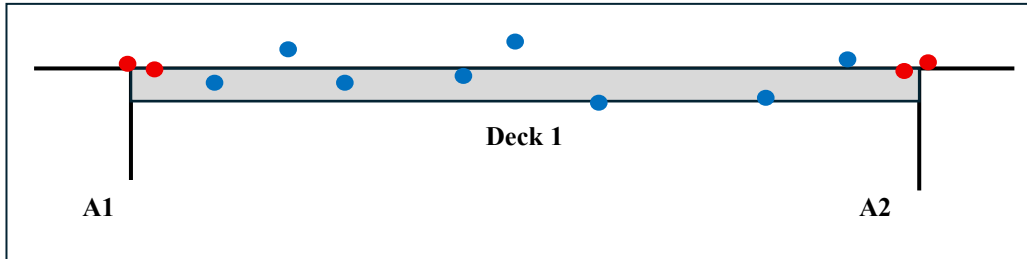
Critical Events & Anomalies:

- **April 2025:** VV improves briefly to 0 mm, while VH remains strongly negative (–14 mm), indicating partial vertical relief without lateral recovery.
- **May 2025:** VH remains between –6 mm and –12 mm, while VV drops again to –12 mm, highlighting continued deformation with no stabilization trend.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits strongly asymmetric vertical displacement, indicating pronounced differential vertical response between the LHS and RHS supports. The maximum vertical uplift is observed at the RHS support region, reaching approximately +16.31 mm, representing the dominant vertical response of the deck.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability across the deck width, reflecting torsional deformation behaviour. The maximum horizontal displacement is observed at the LHS support region, reaching approximately +14.05 mm, indicating dominant lateral movement on the LHS

CONCLUSION:

- The bridge deck slab displays a torsion-dominated deformation response, characterized by large vertical uplift combined with asymmetric horizontal displacement. Overall, the magnitude and asymmetry of the observed displacements suggest structurally significant behaviour, warranting continued monitoring and detailed assessment under varying load and environmental conditions.